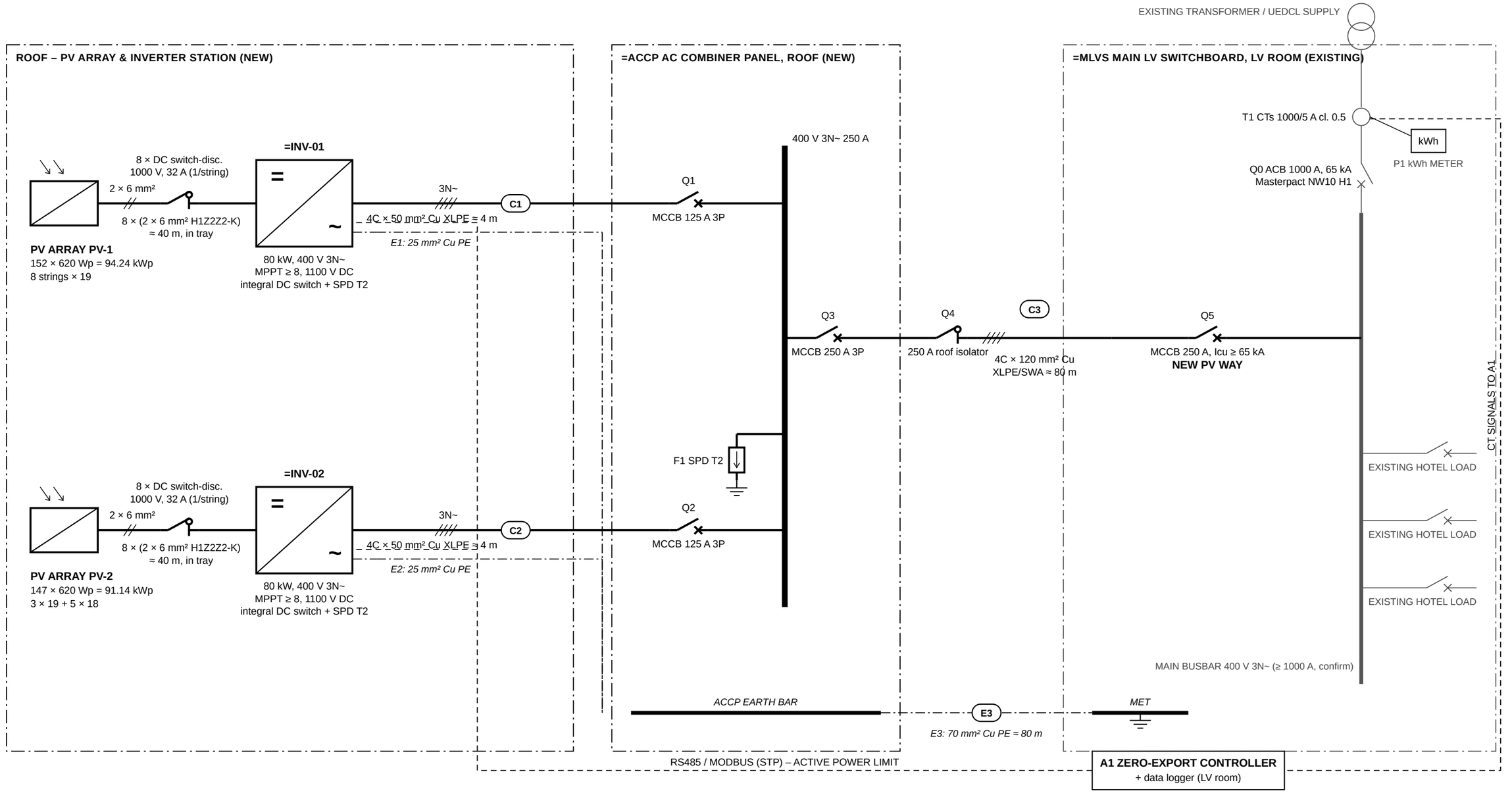


SYMBOLS (IEC 60617)

- PV array / PV generator
- Inverter DC/AC
- Circuit breaker (MCCB / ACB)
- Switch-disconnector (isolator)
- Surge protective device (SPD)
- Current transformer (CT)
- Earth / main earthing terminal
- Heavy = new, light = existing; strokes = conductors
- DC · PE conductor · control / comms

CABLE SCHEDULE

TAG	FROM → TO	CABLE	LENGTH	PROTECTION
S1–S16	PV strings → inverters	2 × 6 mm² Cu H1Z2Z2-K	≈ 40 m	DC switch-disc. 32 A
C1, C2	INV-01/02 → ACCP Q1/Q2	4C × 50 mm² Cu XLPE	≈ 4 m	MCCB 125 A
C3	ACCP Q4 → MLVS Q5	4C × 120 mm² Cu XLPE/SWA	≈ 80 m	MCCB 250 A
E1, E2	INV-01/02 → ACCP earth bar	25 mm² Cu (G/Y)	≈ 4 m	–
E3	ACCP earth bar → MET	70 mm² Cu (G/Y)	≈ 80 m	–
–	Array bonding / spine	16 mm² / 35 mm² Cu (G/Y)	≈ 280 m	–
W1	INV-01 → INV-02 → A1	RS485 STP 2 × 2 × 0.5 mm²	≈ 90 m	–

Lengths indicative – confirm on site. Sizes per IEC 60364-5-52 / 5-54 (see design report).

NOTES

1. 299 × 620 Wp = 185.4 kWp DC; 2 × 80 kW = 160 kW AC; DC/AC 1.16.
2. Grid-tied, zero export, no battery. A1 limits output via CTs T1.
3. New works heavy; existing MLVS equipment light (to be verified).
4. Final ratings to equipment datasheets and detailed design.

CLIENT: **Tororo PV Power Company Limited**	PROJECT: **Kampala Marriott – Rooftop PV 185.4 kWp**

TITLE: **SINGLE-LINE DIAGRAM – PV SYSTEM & GRID CONNECTION**

DRAWING No.	REV	SCALE / SIZE	DATE
PV-SLD-D18102-004	**A**	**NTS / A3**	**05.11.2026**

DRAWN	CHECKED	APPROVED